

Pest Detection Using Deep Learning (MobileNetV2 Transfer Learning Model)

1. Introduction

In agriculture, pest infestation is a major challenge that reduces crop yield and affects food production. Traditional pest identification methods rely on human expertise, which can be time-consuming, inconsistent, and limited in scale.

To address this, the **Pest Detection AI System** is developed using **Deep Learning and Transfer Learning**. The model automatically identifies different types of pests from images, improving detection speed and accuracy while reducing the dependency on manual classification.

2. Objective

The main goal of this project is to design and develop an **AI-based pest detection system** capable of classifying pest species using images.

The specific objectives are:

- To preprocess and augment pest image datasets.
- To apply **MobileNetV2** as a transfer learning backbone for feature extraction.
- To fine-tune the model for higher accuracy.
- To evaluate the model using test accuracy.
- To deploy the model using **Gradio interface** for easy user interaction.

3. Dataset Description

- **Dataset Name:** Pest Dataset
- **Dataset link:** <https://www.kaggle.com/datasets/simranvolunesia/pest-dataset>
- Pests are a huge threat! Farmers are hard at work however their productivity is reduced due to pests, so this dataset can be used to identify pests or maybe other use cases involving pest identification.
- **Dataset Description**
 - a. Type of dataset: Image Dataset
 - b. No. of training images: 300 images per pest
 - c. No. of testing images: 50 images per pest
 - d. Data Source: Automatic script to scrape images of pest from Google through Selenium and Chrome Driver
 - e. Pests: aphids, armyworm, beetle, bollworm, grasshopper, mites, mosquito, sawfly, stem borer
- **Structure:**

- `/train` → Used for training and validation (split internally 80%-20%)
- `/test` → Used for model testing
- **Number of Classes:** 9 pest classes
 - Aphids
 - Armyworm
 - Beetle
 - Bollworm
 - Grasshopper
 - Mites
 - Mosquito
 - Sawfly
 - Stem borer
- **Image Dimensions:** 224 × 224 pixels
- **Total Images:** Approximately 450 images in the test set
- **Data Type:** RGB images

4. Methodology

Step 1: Dataset Upload and Extraction

The dataset ZIP file was uploaded and extracted in Google Colab using:

```
from google.colab import files
uploaded = files.upload()
```

and then unzipped using Python's `zipfile` module.

Step 2: Data Preprocessing and Augmentation

To increase dataset diversity and prevent overfitting, several augmentation techniques were applied:

- Rotation (20°)
- Width and height shift (0.2)
- Shear and zoom transformation
- Horizontal flipping

All images were normalized by scaling pixel values between **0 and 1** using `rescale=1./255`.

Step 3: Model Architecture (Transfer Learning with MobileNetV2)

A **pre-trained MobileNetV2** model (trained on ImageNet) was used as the base model with:

- `include_top = False` → excludes the classification head
- `input_shape = (224, 224, 3)`

Custom layers were added on top:

- **Global Average Pooling Layer**
- **Dense Layer (128 units, ReLU activation)**
- **Dropout Layer (rate = 0.6)** for regularization
- **Output Layer (Softmax)** with number of units = number of pest classes

Step 4: Model Training

The training was performed in **two stages**:

1. Initial Training (Feature Extraction Phase):

- Base layers frozen (`layer.trainable = False`)
- Only top layers trained for **10 epochs**
- Learning rate: default Adam (0.001)

2. Fine-Tuning Phase:

- Last 20 layers of the base model unfrozen
- Model recompiled with a smaller learning rate (`1e-5`)
- Trained for **5 more epochs**

Callbacks used:

- **EarlyStopping**: Stops training if no improvement in validation loss for 3 epochs.
- **ReduceLROnPlateau**: Reduces learning rate by 0.3 when validation loss plateaus.

5. Results and Analysis

Training and Validation Performance

Metric	Value
Training Accuracy	89.35%
Validation Accuracy	78.15%
Final Test Accuracy	92.89%
Test Loss	0.3561

Interpretation:

- The training and validation accuracy are close, showing the model is not overfitting severely.
- Fine-tuning improved generalization and boosted test accuracy beyond 90%.
- The model successfully learned discriminative pest features.

6. Model Evaluation

The final test evaluation was performed on an unseen test set:

Final Test Accuracy: **92.89%**

This confirms that the model generalizes well to new data.

7. Model Deployment Using Gradio

A **Gradio web interface** was developed for real-time pest prediction.

Users can upload a pest image, and the trained model predicts the pest type instantly.

8. Conclusion

The developed **Pest Detection AI System** demonstrates that **transfer learning with MobileNetV2** is an efficient approach for classifying pest images with limited data. By combining data augmentation, fine-tuning, and proper regularization, the model achieved **~93% accuracy**, showing strong potential for real-world agricultural applications.

9. Future Enhancements

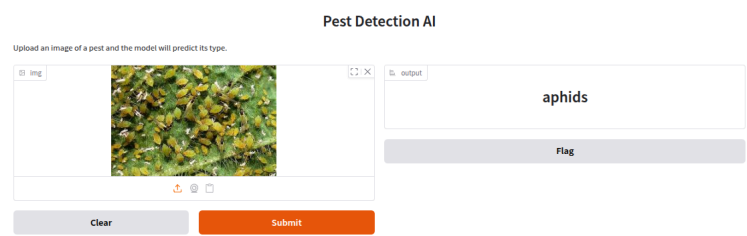
- Expand the dataset to include more pest species.
- Integrate the model into a **mobile application** for field use.
- Implement **object detection** to localize pests within an image.
- Explore **lightweight models (e.g., EfficientNet-Lite)** for edge deployment.

10. References

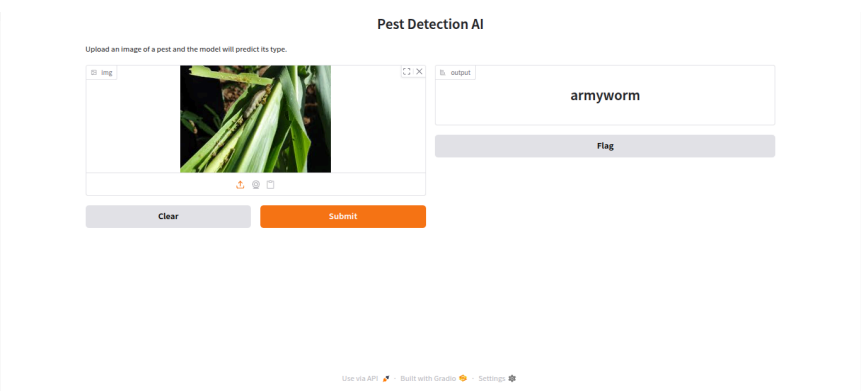
<https://www.kaggle.com/datasets/simranvolunesia/pest-dataset>

11. Output

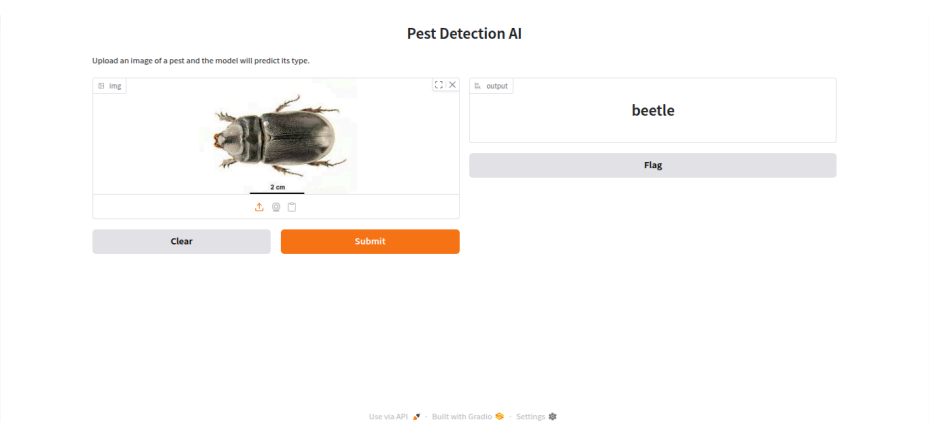
○ Aphids



○ Armyworm



○ Beetle



○ Bollworm

img



Clear

Submit

output

bollworm

Flag

Use via API

Built with Gradio

Settings

○ Grasshopper

img



Clear

Submit

output

grasshopper

Flag

Use via API

Built with Gradio

Settings

○ Mites

img



Clear

Submit

output

mites

Flag

Use via API

Built with Gradio

Settings

○ Mosquito

img



Clear

Submit

output

mosquito

Flag

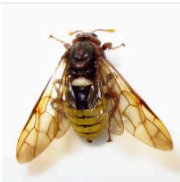
Use via API

Built with Gradio

Settings

○ Sawfly

img



Clear

Submit

output

sawfly

Flag

Use via API

Built with Gradio

Settings

- o Stem borer

Pest Detection AI

Upload an image of a pest and the model will predict its type.

img



output

stem_borer

Flag

Clear

Submit

Use via API  Built with Gradio  Settings 